

OADP Deployment, configuration, restoration, backup and scheduling

Objective

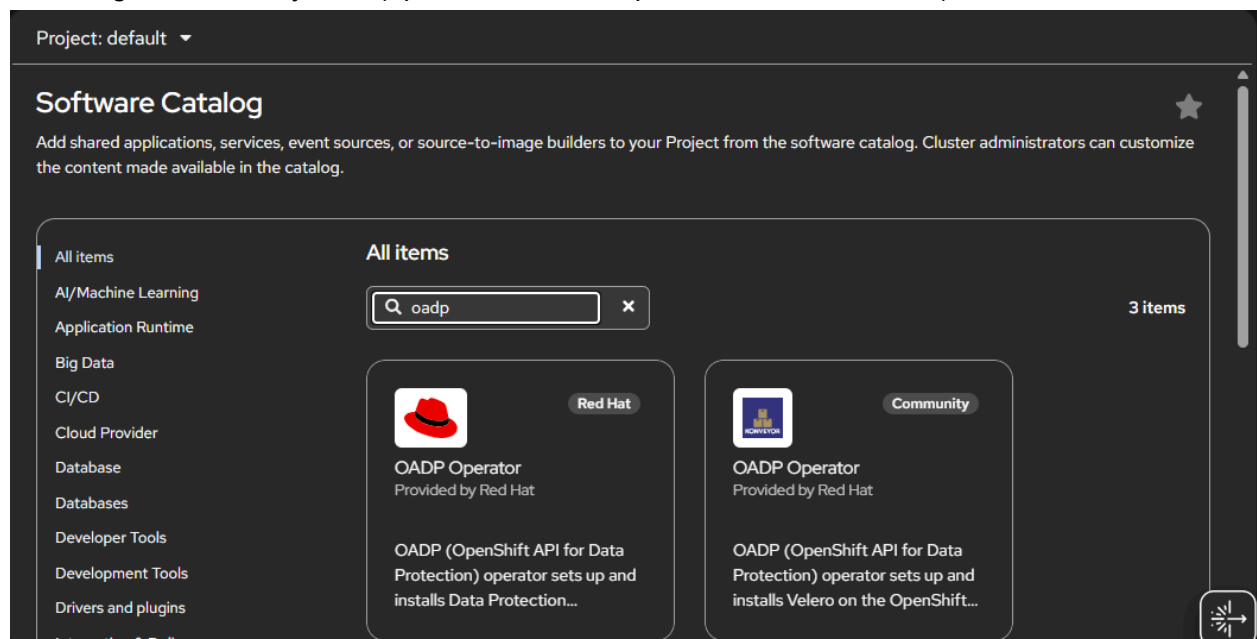
OpenShift API for Data Protection (OADP) is an operator that enables backup, restore, and disaster recovery of applications and cluster resources in an OpenShift environment using Velero. In this implementation, OADP is deployed and configured to protect workloads by taking backups of namespaces, persistent volumes, and application data, storing them in an object storage location. The setup includes performing on-demand backups, restoring applications from those backups to validate recovery, and configuring scheduled backups to automate data protection. This ensures that the cluster and its applications can be reliably recovered in case of failures or data loss.

Prerequisites

- OADP operator images mirrored
- Cluster access
- Sample application for testing
- Object storage

Operator installation

S1: navigate to Ecosystem (operator hub for ocp cluster which is >4.20) search OADP



Install with default configuration

Check in “openshift-adp” namespace if the following resources came up

```
[lab-user@bastion ~]$ oc get all
Warning: apps.openshift.io/v1 DeploymentConfig is deprecated in v4.14+, unavailable in v4.10000+
NAME                                READY   STATUS    RESTARTS   AGE
pod/node-agent-n7752                1/1     Running   0           51m
pod/node-agent-tnh47                1/1     Running   0           51m
pod/node-agent-vrhxf                1/1     Running   0           51m
pod/openshift-adp-controller-manager-7d5c979b6c-c4wc2  1/1     Running   1           15h
pod/velero-547ffcbfc-g4v5d          1/1     Running   0           35m

NAME                                TYPE             CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
service/openshift-adp-controller-manager-metrics-service  ClusterIP        172.31.208.47    <none>           8443/TCP         15h
service/openshift-adp-velero-metrics-svc                  ClusterIP        172.31.5.229     <none>           8085/TCP         51m

NAME                DESIRED   CURRENT   READY   UP-TO-DATE   AVAILABLE   NODE SELECTOR          AGE
daemonset.apps/node-agent  3         3         3       3             3           kubernetes.io/os=linux  51m

NAME                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/openshift-adp-controller-manager  1/1     1             1           15h
deployment.apps/velero                            1/1     1             1           51m

NAME                DESIRED   CURRENT   READY   AGE
replicaset.apps/openshift-adp-controller-manager-7d5c979b6c  1         1         1       15h
replicaset.apps/velero-547ffcbfc                            1         1         1       51m
[lab-user@bastion ~]$
```

Sample application deployment

Deploy the following sample application for testing.

```
apiVersion: v1
kind: Namespace
metadata:
  name: oadp-demo
---
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: postgres-pvc
  namespace: oadp-demo
spec:
  accessModes:
    - ReadWriteOnce
  storageClassName: ocs-external-storagecluster-ceph-rbd
  resources:
    requests:
      storage: 1Gi
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: postgres
  namespace: oadp-demo
spec:
```

```
replicas: 1
selector:
  matchLabels:
    app: postgres
template:
  metadata:
    labels:
      app: postgres
  spec:
    containers:
      - name: postgres
        image: registry.redhat.io/rhel8/postgresql-13
        env:
          - name: POSTGRESQL_USER
            value: testuser
          - name: POSTGRESQL_PASSWORD
            value: testpass
          - name: POSTGRESQL_DATABASE
            value: testdb
        ports:
          - containerPort: 5432
        volumeMounts:
          - name: data
            mountPath: /var/lib/pgsql/data
    volumes:
      - name: data
        persistentVolumeClaim:
          claimName: postgres-pvc
---
apiVersion: v1
kind: Service
metadata:
  name: postgres
  namespace: oadp-demo
spec:
  selector:
    app: postgres
  ports:
    - port: 5432
```

Backup

We need a Object storage to store our backups as OADP uses only object storage to store backup

S1: Create a ObjectBucketClaim and get it's endpoint, bucket name, access key, access secret

```
[lab-user@bastion ~]$ oc get sc
NAME                                     PROVISIONER
RECLAIMPOLICY  VOLUMEBINDINGMODE  ALLOWVOLUMEEXPANSION  AGE
ocs-external-storagecluster-ceph-rbd
openshift-storage.rbd.csi.ceph.com      Delete              Immediate
true                                     20h
ocs-external-storagecluster-ceph-rbd-immediate
openshift-storage.rbd.csi.ceph.com      Delete              Immediate
true                                     20h
ocs-external-storagecluster-cephfs
openshift-storage.cephfs.csi.ceph.com   Delete              Immediate
true                                     20h
openshift-storage.noobaa.io
openshift-storage.noobaa.io/obc         Delete              Immediate
false                                    20h
[lab-user@bastion ~]$
```

We have noobaa storage (ODF) which provides us object storage
Create Object Bucket Claim by applying the following

```
apiVersion: objectbucket.io/v1alpha1
kind: ObjectBucketClaim
metadata:
  name: velero-backup
  namespace: openshift-adp
spec:
  generateBucketName: velero-backup
  storageClassName: openshift-storage.noobaa.io
```

It should be bound

```
[lab-user@bastion ~]$ oc get objectbucket
NAME                                     STORAGE-CLASS
CLAIM-NAMESPACE  CLAIM-NAME  RECLAIM-POLICY  PHASE  AGE
obc-openshift-adp-velero-backup  openshift-storage.noobaa.io
Delete              Bound      61m
[lab-user@bastion ~]$
```

Bucket name endpoint (bucket host)

```
[lab-user@bastion ~]$ oc extract --to= cm/velero-backup
# BUCKET_HOST
s3.openshift-storage.svc
# BUCKET_NAME
velero-backup-1eb43169-6788-4358-a143-4b75de33741b
# BUCKET_PORT
443
# BUCKET_REGION

# BUCKET_SUBREGION

[lab-user@bastion ~]$
```

AWS Access key ID and AWS secret access key

```
[lab-user@bastion ~]$ oc extract --to - secret/velero-backup
# AWS_ACCESS_KEY_ID
ywAIOn1g5wHNNJiwthE5
# AWS_SECRET_ACCESS_KEY
cJX44ebKDIPB7fsB76dFi47VIwMdpyteHnqtkIs6
[lab-user@bastion ~]$
```

S2: create DataProtectionApplication by using details you received from previous step

```
apiVersion: oadp.openshift.io/v1alpha1
kind: DataProtectionApplication
metadata:
  name: velero-sample
  namespace: openshift-adp
spec:
  configuration:
    velero:
      defaultPlugins:
        - openshift
        - aws
        - csi
      featureFlags:
        - EnableCSI
    nodeAgent:
      enable: true
```

```

    uploaderType: kopia
  backupLocations:
  - name: default
    velero:
      provider: aws
      default: true
      config:
        profile: "default"
        region: "noobaa"
        s3Url: "https://s3.openshift-storage.svc:443"
        s3ForcePathStyle: "true"
        insecureSkipTLSVerify: "true"
      credential:
        name: cloud-credentials
        key: cloud
      objectStorage:
        bucket: "velero-backup-1eb43169-6788-4358-a143-4b75de33741b"
        prefix: velero

```

Verify

```

[lab-user@bastion ~]$ oc get dpa
NAME              RECONCILED  AGE
velero-sample    True        66m
[lab-user@bastion ~]$

```

S3: create a backup custom resource

```

apiVersion: velero.io/v1
kind: Backup
metadata:
  name: postgres-backup
  namespace: openshift-adp
spec:
  includedNamespaces:
  - oadp-demo
  storageLocation: default
  ttl: 24h

```

[NOTE: We can backup Specific resources by mentioning them in “includedResources:”]

Example:

```
apiVersion: velero.io/v1
kind: Backup
metadata:
  name: postgres-pvc-backup
  namespace: openshift-adp
spec:
  includedNamespaces:
    - oadp-demo
  includedResources:
    - persistentvolumeclaims
  snapshotVolumes: true
  storageLocation: default
  ttl: 24h
```

Verify

```
[lab-user@bastion ~]$ oc get backups.velero.io
NAME                                AGE
postgres-backup                    49m
[lab-user@bastion ~]$
```

Restore

Create a restore Custom resource

```
apiVersion: velero.io/v1
kind: Restore
metadata:
  name: postgres-restore
  namespace: openshift-adp
spec:
  backupName: postgres-backup
```

[NOTE: We can restore the application in different namespace as well]

Example :

```
apiVersion: velero.io/v1
kind: Restore
metadata:
  name: postgres-restore-new-namespace
  namespace: openshift-adp
```

```
spec:
  backupName: postgres-backup
  namespaceMapping:
    oadp-demo: Application-restored
```

[NOTE: We can restore specific resource from backup as well, by mentioning them in `includedResources`]

Example:

```
apiVersion: velero.io/v1
kind: Restore
metadata:
  name: postgres-pvc-restore
  namespace: openshift-adp
spec:
  backupName: postgres-backup
  includedResources:
    - persistentvolumeclaims
```

Verify

```
[lab-user@bastion ~]$ oc get restores.velero.io
NAME              AGE
postgres-restore  27m
[lab-user@bastion ~]$
```

```
[lab-user@bastion ~]$ oc describe restores.velero.io/postgres-restore
```

<<<<<<OUTPUT OMITTED >>>>>>

```
Status:
  Completion Timestamp: 2026-07-10T06:00:36Z
  Hook Status:
  Phase: Completed
  Progress:
    Items Restored: 34
    Total Items: 34
  Start Timestamp: 2026-07-10T06:00:33Z
  Warnings: 4
  Events: <none>
```

Phase should be completed

Schedule For backup

Create a scheduler Custom resource

```
apiVersion: velero.io/v1
kind: Schedule
metadata:
  name: postgres-backup-schedule
  namespace: openshift-adp
spec:
  schedule: "0 */6 * * *"
  template:
    includedNamespaces:
      - oadp-demo
    storageLocation: default
    ttl: 24h
```

Verify

```
[lab-user@bastion ~]$ oc get schedule
NAME                                STATUS    SCHEDULE           LASTBACKUP    AGE
PAUSED
postgres-backup-schedule           Enabled   0 */6 * * *        32m           41m
[lab-user@bastion ~]$
```

```
[lab-user@bastion ~]$ oc get backups.velero.io
NAME                                AGE
postgres-backup                    56m
postgres-backup-schedule-20260710055540 38m
postgres-backup-schedule-20260710060040 33m
[lab-user@bastion ~]$
```